MICHAEL F. PÉREZ

ML & Computer Vision Engineer · Ph.D. in Computer Science · Tampa, FL (open to relocation)
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ML and computer vision engineer (Ph.D. in Computer Science) with experience integrating computer vision into production software, building ML data and evaluation workflows, and researching video understanding.

TECHNICAL SKILLS

ML & computer vision: PyTorch, TensorFlow, OpenCV, scikit-learn, Hugging Face, VLMs (LaViLa, LLaVA, Qwen)
MLOps & data tools: Kubernetes, Airflow, MLflow, FiftyOne, Label Studio, Docker, SLURM/HPC
Languages & systems: Python, C++, SQL, JavaScript, FastAPI, Flask, Celery, Redis, Linux, Git, NumPy, pandas, SciPy

EXPERIENCE

Senior Engineer, Raft · Tampa, FL — Aug 2026 – Nov 2026

- Developed and integrated computer vision data exploration and analysis capabilities into a larger production software system, including dataset visualization, clustering, filtering, natural-language querying, and geospatial analytics.
- Built ML data management and evaluation workflows for dataset curation, annotation review, model analysis, and debugging.
- Contributed to the development, testing, and integration of AI/ML capabilities, including object-detection experimentation and improvements to existing software systems.

Machine Learning Engineer Intern, CoVar · Durham, NC — May 2025 – Aug 2025

- Applied infrared video super-resolution to long-range object classification; first-author paper at the 2026 Military Sensing Symposium (MSS).
- Trained lightweight RGB object-detection models targeting real-time, on-device drone detection.
- Built offline computer vision evaluation pipelines spanning preprocessing, inference, metrics, and efficiency analysis.

Graduate Research Assistant, University of Florida · Gainesville, FL — Aug 2020 – Aug 2026

- Ph.D. thesis: designed a causal, online temporal action-detection model on state-space (Mamba) backbones that matches the strongest verified strictly-causal detector on THUMOS at roughly 60% of its parameters and 2.5–2.9× its throughput, running in real time at 37.7 FPS.
- Built UI and image/video analysis APIs and trained action-segmentation models for CReLeRI, a DARPA-funded explainable long-video analysis system; first author of its ACM Multimedia 2025 demo.
- Built CReLeRI's long-video analysis web tool: FastAPI backend with Celery and Redis background tasks, containerized with Docker, with GPU compute offloaded to UF's HiPerGator cluster.
- Trained and tested action-recognition models, including fine-tuning the video-language model LaViLa, for ENKIx, a DARPA-funded real-time AR task-guidance system, and ran usability and cognitive-load evaluations.

EDUCATION

Ph.D. in Computer Science, University of Florida — Aug 2026
M.S. in Computer Science, University of Florida — May 2023
B.S. in Computer Science, Florida Polytechnic University — May 2020

SELECTED PUBLICATIONS

CReLeRI: Explainable, Concept-centric, Representation, Learning, Reasoning, and Interaction Video Analysis System. First author · ACM Multimedia 2025 (demo) · doi:10.1145/3746027.3754479

Super-Resolution for Improving Vehicle Recognition at Large Stand-offs. First author · MSS 2026

MuCHEx: A Multimodal Conversational Debugging Tool for Interactive Visual Exploration of Hierarchical Object Classification. Co-author · IEEE Computer Graphics and Applications, 2025 · doi:10.1109/MCG.2025.3598204

CERTIFICATION, HONORS & SERVICE

Certification: CompTIA Security+ (2026)
Honors: L3Harris Graduate Fellowship (one of five UF CISE recipients, 2022); UF Graduate School Preeminence Award
Service: Peer reviewer for ACM Multimedia, ACM ICMR, IEEE VR, and ICXR (2024 – 2026)